Clément MAGENTIES

Engineering Student · UAV Flight Dynamics & Embedded Systems
Toulouse INP-ENSEEIHT · Major in Electronics, Electrical Energy & Automation (3EA)
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linkedin.com/in/magenties-clement · **Certified Drone Pilot (EU/EASA A1/A3 since 2020)**



Profile

Engineering student at **ENSEEIHT** specializing in **Automatic Control, Embedded Systems, and Signal Processing**. Passionate about drone robotics with hands-on expertise in custom FPV quadcopters, 2 kHz Blackbox telemetry parsing, frequency-domain identification (FFT), and low-level firmware in C/FPGA. Seeking a **2026/2027 internship or junior engineering position in UAV flight control, embedded software, or flight testing in Switzerland**.

Technical Skills

- **Flight Control & UAV Systems:** PID loop tuning, state estimation, biquad & dynamic notch filtering, 2 kHz Blackbox telemetry parsing, FFT spectral analysis, Betaflight, Gyroflow, ExpressLRS (2.4 GHz), FPV piloting.
- **Embedded Systems & Architecture:** Low-level C, bare-metal development, real-time interrupts (preemptive & event-driven ISRs), softcore NIOS II (32-bit RISC), Intel Cyclone V FPGA, Avalon bus, memory-mapped I/O.
- **Hardware, Electronics & Sensors:** PWM / 4-quadrant H-bridge motor drives, sensor interfacing (IMUs, SRF05 ultrasonic time-of-flight, optical line sensors), digital oscilloscopes, logic analyzers, PCB bring-up & soldering.
- **Scientific Software & Tools:** Python (`NumPy`, `SciPy`, `Matplotlib`), Quartus Prime, Eclipse NIOS II, Git, Linux, Latis Pro.
- **Certifications & Licensing: Certified Drone Pilot (EU/EASA sub-categories A1 & A3 since 2020)**, French Driving License (Permis B).

Engineering & UAV Projects

Custom 6-Inch FPV Quadcopter — Flight Dynamics & Control Loop Tuning *2025 – Present*
Personal Engineering Project · Python, Betaflight, Blackbox Telemetry, PID Control, Biquad Filtering

- Designed, wired, and assembled a custom 6-inch quadcopter optimized for high-speed tracking and cinematic flight authority.
- Recorded and decoded 2 kHz onboard Blackbox flash logs using custom Python scripts to compare pilot setpoint against gyroscopic angular rate response across Roll, Pitch, and Yaw axes.
- Conducted FFT spectral analysis isolating a critical 180 Hz airframe resonance; implemented motor RPM telemetry-synced dynamic notch and biquad low-pass filters to prevent derivative (D-term) noise saturation.
- Fine-tuned proportional (P) and derivative (D) gains, eliminating propwash oscillations and motor heat spikes under heavy dynamic load.

Mobile Robot "BoB" — Hardware/Software Co-Design & Embedded Control *ENSEEIHT · 2024 – 2025*
Embedded Systems Project · Low-Level C, Intel Cyclone V FPGA, 32-bit NIOS II Softcore, Quartus Prime

- Synthesized combinational address decoding logic in Quartus Prime to map a 64-byte address space into 8 peripheral select signals (`CS0`–`CS7`) on an Intel Cyclone V FPGA (Terasic DE1-SoC).
- Authored bare-metal C device drivers for 7-segment displays, LEDs, 3-channel optical ground tracking sensors, and 3 SRF05 ultrasonic rangefinders (0–100 cm) via memory-mapped volatile pointers and bitwise masking.
- Implemented 4-quadrant PWM motor control (10 ms astable base timer, monostable pulse width modulation from 1 ms to 2 ms).
- Designed prioritized preemptive multi-timer ISRs and aperiodic event-driven interrupts using a hardware D flip-flop state machine, ensuring instant obstacle avoidance and emergency braking (< 20 cm).

Flywheel Energy Storage System (FESS) — Modeling & Dynamic Testing *CPGE Pierre de Fermat · 2024 – 2025*
Preparatory Project (TIPE) · Python (NumPy, SciPy), Electromechanics, Signal Processing

- Engineered an experimental test bench coupling a DC machine to interchangeable rotors to study grid frequency stabilization.
- Characterized rotor moment of inertia J via torsion pendulum dynamics ($J = \frac{CT^2}{4\pi^2}$); filtered optical sensor data using cubic splines (`CubicSpline`) and integrated power curves (`np.trapz`) to evaluate round-trip efficiency (concrete cylinder vs. bicycle wheel).

Education

Toulouse INP-ENSEEIHT — Toulouse, France *2025 – 2028*
Diplôme d'Ingénieur (Master of Science in Engineering) · Major in Electronics, Electrical Energy & Automation (3EA)
Core Curriculum: Control Theory, Real-Time Embedded Systems, Signal Processing, Power Electronics, Computer Architecture.

Lycée Pierre de Fermat — Toulouse, France *2023 – 2025*
CPGE PCSI / PC (Physics & Chemistry) · Intensive two-year undergraduate preparation in Mathematics, Physics & Engineering.

Lycée Marie-Curie — Tarbes, France *2020 – 2023*
French Baccalauréat · Highest Honors (*Mention Très Bien* with congratulations of the jury).

Leadership, Languages & Interests

- **Founder & President, TEAMNEPAL Non-Profit (2025 – Present):** Led a 5-week field mission at Victor Hugo School (Chapali, Nepal); secured 10,800 € funding (FONJEP, FSDIE); organized school Olympics for 260 pupils and co-created 3 short films.
- **Languages: French** (Native), **English** (Fluent / C1, full professional working proficiency), **German** (A2 certified — *Deutschesprachdiplom*, asset for Swiss-German cantons including Zurich).
- **Interests & Teamwork:** Regional competition Handball (10+ years, team tactics & quick decision-making under pressure), Road cycling, Alpine mountaineering & trekking (Pyrenees, Himalayas), Video post-production (TVN7, DaVinci Resolve).